Mini Project – Electric Piano

Author: Devesh Maheshwari
Section: 22
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Abstract

This project explores the functionality and applications of the 555 timer integrated circuit through two key experiments: a comparative study of its monostable and astable modes, and a real-world application in the form of an electric piano. The monostable circuit was designed to generate a 3-second high output pulse, while astable configurations with 60% and 75% duty cycles demonstrated continuous pulse generation. The electric piano application used variable resistor networks to control the output frequency, producing different musical tones through a speaker. Simulations and oscilloscope readings verified circuit behavior, with minor deviations due to component tolerances and user error. The study highlights the flexibility of the 555 timer in timing and waveform generation, while the piano project shows its use in audio projects.

Link: https://youtu.be/6dxMKBkFDPA

Introduction

The 555 timer was invented in 1971. It was designed to be a simple, versatile, and cost-effective timing solution for use in various electronics applications. The chip quickly gained popularity due to its reliability and ease of use and has since become one of the most widely used integrated circuits in history.
The 555 timer is commonly used in monostable mode for applications like timer switches, pulse generators, and debouncing circuits. In astable mode, it acts as an oscillator for LED blinkers, PWM control for motors. Its simplicity and versatility also make it useful in touch sensors, frequency converters etc.

In monostable mode, the 555 timer acts as a one-shot pulse generator. When a trigger signal is applied (low voltage), the output goes high for a specific duration determined by an external resistor and capacitor ($T = 1.1 \times R \times C$). After this time, the output returns to low until it is triggered again.

In astable mode, the 555 timer operates as a continuous oscillator. It continuously switches between high and low states without requiring any external trigger. The timing of the high (Ton) and low (Toff) periods is set using two resistors and a capacitor, allowing control over the frequency and duty cycle of the output waveform.

Objectives

The overall objective of this project is to explore the functionality, configurations, and practical applications of the 555 Timer integrated circuit.

Experiment A – Comparative Study of 555 Timer Modes

The objective of this experiment is to:

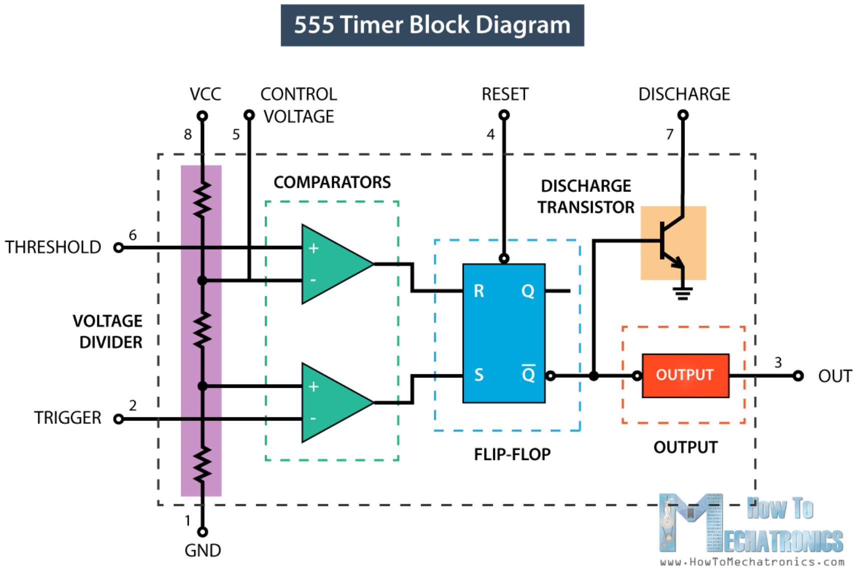
- Understand the internal operation and pin functions of the 555 timer using the datasheet as a reference.
- Design and construct circuits that demonstrate the monostable and astable configurations of the 555 timer.
- Observe and analyze the timing behavior of the output signal in both configurations.

Experiment B – Application of 555 Timer

- Select a real-world application of the 555 timer that aligns with personal or academic interests.
- Design and build a functional circuit using the timer for the chosen application
- Simulate the circuit behavior using SPICE and validate it with oscilloscope measurements.

Theory

555 Timer Overview



555 Timer Block diagram (Dejan)

The 555 timer consists of three main parts: a voltage divider with comparators, a flip-flop, and a discharge/output stage.

Three resistors form a voltage divider that provides reference levels at 1/3 Vcc and 2/3 Vcc. These levels are compared with the voltages at the Trigger (Pin 2) and Threshold (Pin 6). When the Trigger drops below 1/3 Vcc, the lower comparator sets the flip-flop, making the Output (Pin 3) go high. When the Threshold exceeds 2/3 Vcc, the upper comparator resets the flip-flop, making the output go low.

The flip-flop controls a discharge transistor connected to Pin 7, which discharges the timing capacitor when on. The Control Voltage (Pin 5) can adjust the threshold level, while Reset (Pin 4) can override the flip-flop when pulled low.

In monostable mode, the timer outputs one pulse per trigger. In astable mode, it continuously oscillates, producing a square wave output.

Monostable mode of operation

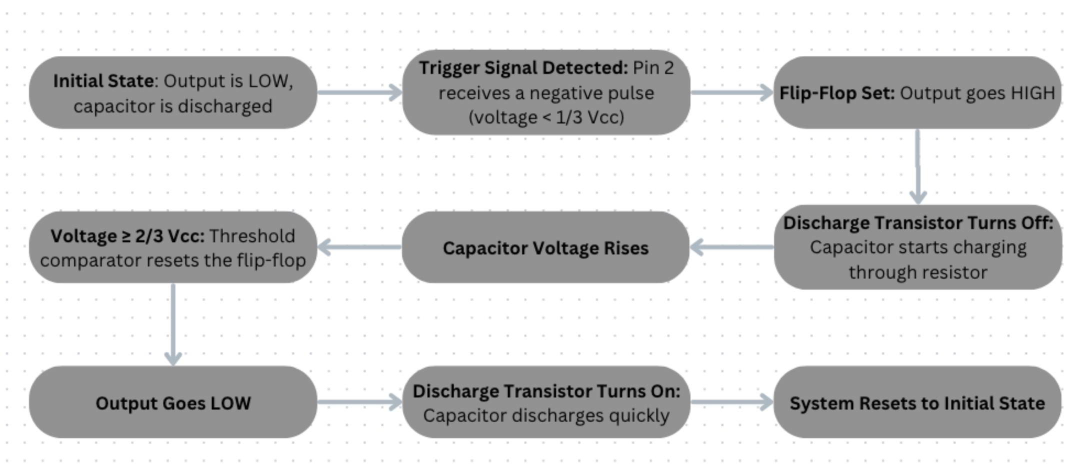
In monostable mode, the 555 timer functions as a one-shot pulse generator—it produces a single output pulse of a fixed duration in response to a trigger signal.

Initially, the Output (Pin 3) is low, and the capacitor connected to the Discharge pin (Pin 7) is held at ground by the internal transistor. When a negative trigger pulse is applied to the Trigger pin (Pin 2) and the voltage drops below 1/3 Vcc, the lower comparator sets the flip-flop, turning the Output high and switching off the discharge transistor.

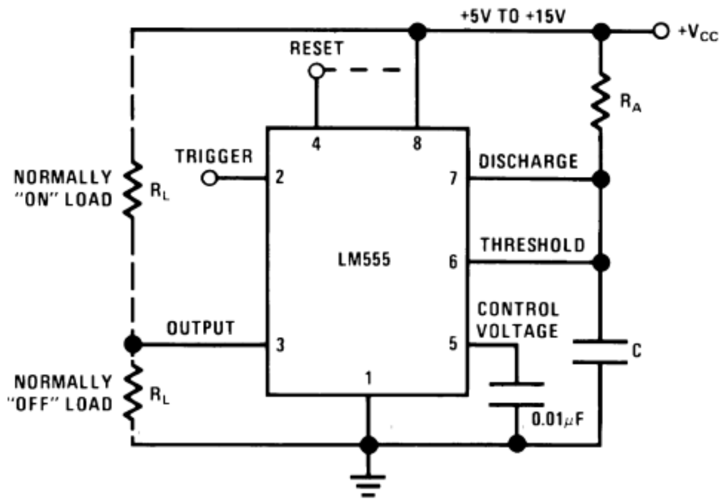
This allows the capacitor to start charging through an external resistor. The duration of the output pulse depends on the R-C time constant. As the capacitor charges, its voltage increases. Once it reaches 2/3 Vcc, the upper comparator resets the flip-flop, turning the output low again and activating the discharge transistor, which quickly discharges the capacitor.

The timer remains in this state until the next trigger pulse is applied. This mode is useful for timers, delay circuits, and pulse generation where a fixed output duration is required.

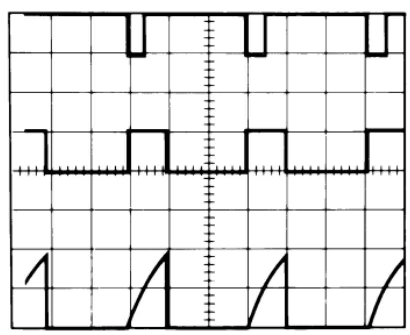
The pulse width is the duration for which the 555 timer's output remains high after being triggered in monostable mode. Pulse width $T = 1.1(RC)$



Monostable Flow Diagram



Monostable General Circuit (Texas Instruments 9)



Top Trace: Input 5V/Div.
Middle Trace: Output 5V/Div.
Bottom Trace: Capacitor Voltage 2V/Div.

Monostable Timing Diagram (Texas Instruments 9)

Astable mode of operation

In astable mode, the 555 timer functions as an oscillator, producing a continuous square wave without any external input. The circuit uses two resistors (Ra and Rb) and a capacitor (C) to set the timing for charging and discharging cycles.

When powered, the capacitor starts charging through both Ra and Rb. As the voltage across the capacitor rises and reaches 2/3 Vcc, the upper comparator resets the internal flip-flop, causing the output (Pin 3) to go low and enabling the discharge transistor (connected to Pin 7) to ground the capacitor through Rb.

Once the capacitor discharges to 1/3 Vcc, the lower comparator sets the flip-flop, turning the output high again and restarting the charging process. This cycle repeats indefinitely, producing a square wave at the output.

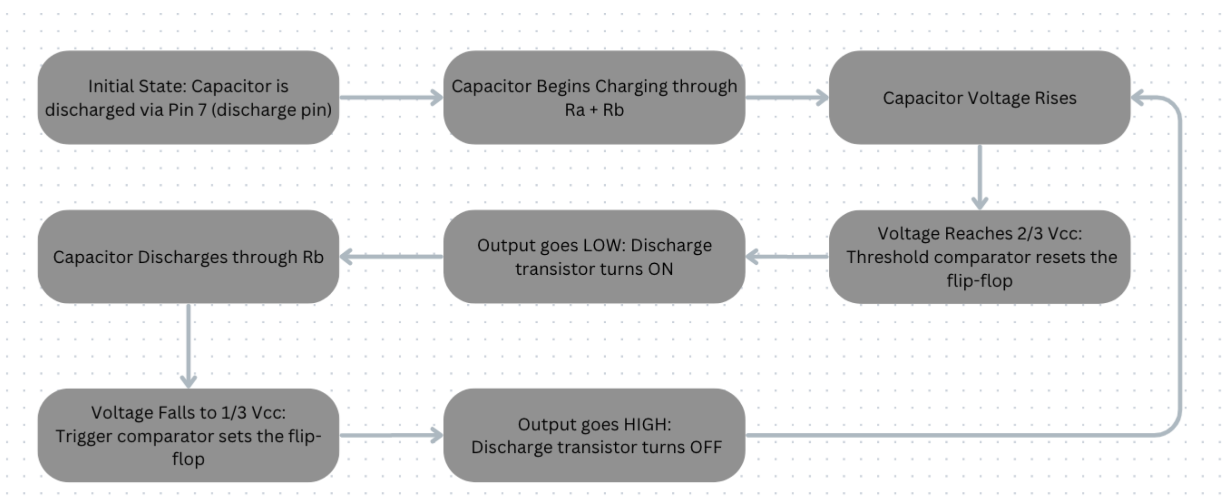
Charge time (output high):
`t1 = 0.693 × (Ra + Rb) × C`

Discharge time (output low):
`t2 = 0.693 × Rb × C`

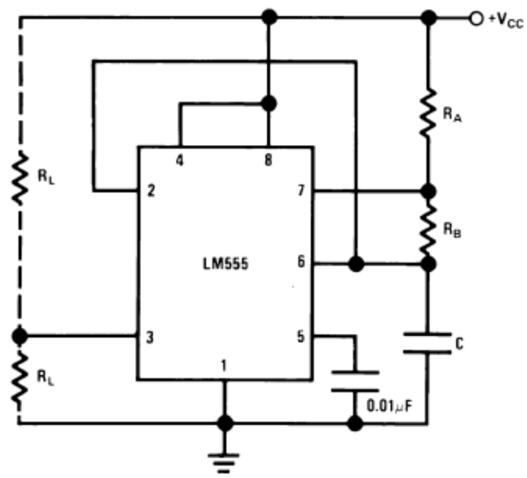
Total period:
`T = t1 + t2 = 0.693 × (Ra + 2Rb) × C`

Frequency of oscillation:
`f = 1 / T = 1.44 / ((Ra + 2Rb) × C)`

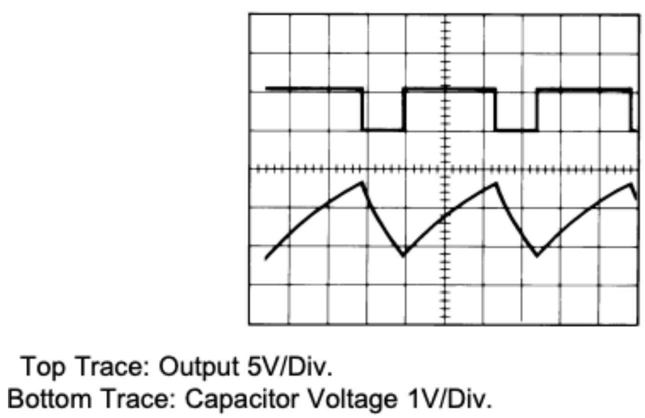
Duty cycle (D):
`D = Ton / (Ton + Toff) = (Ra + Rb) / (Ra + 2 × Rb)`



Astable Flow Diagram



Astable General Circuit (Texas Instruments 10)



Astable Timing Diagram (Texas Instruments 11)

Experimental Procedure

Monostable Mode

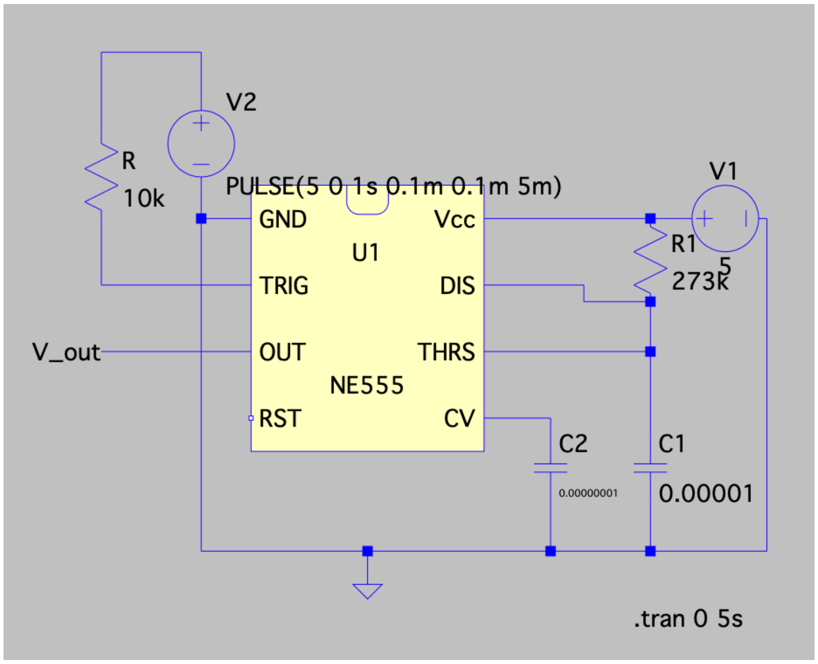


Fig1: Monostable Schematic

- The aim was to create a monostable circuit that has a high output pulse of 3 seconds.
- A capacitor of 10 micro-F and LM555 Timer were chosen for the task
- The required component values for the circuit were calculated as follows
  - T = 1.1 × R × C
    T = 3 seconds
    C = 10 microfarads = 10 × 10^{-6} F
  - 3 = 1.1 × R × (10 × 10^{-6})
  - R = 3 / (1.1 × 10 × 10^{-6})
    R = 3 / (11 × 10^{-6})
    R = (3 / 11) × 10^{6}
    R ≈ 0.2727 × 10^{6}
    R ≈ 272,727 ohms
- The required circuit was constructed in LT Spice and the output value at V_out was graphed for testing.
- The calculated resistance was 273k ohms. The nearest resistance values in the kit allowed a resistance of 220k + 56k = 276k ohms.
- The circuit in the figure 1 was constructed with a 5V power supply. Note that V2 was not part of the actual circuit and was instead replaced with a switch that connected to ground when toggled.
- The trigger pin was directly connected to V1.
- An oscilloscope was attached across the output pin of the timer and the common ground.

- To test, the switch was toggled once, redirecting the power supply to ground and thus triggering the timer.
- The output data was observed on the oscilloscope.

Question: When looking at the datasheet, you will notice that there is a "normally on" and a "normally off" location for the output. What do both of these mean? Why would one be able to be on when the other is off?

In the 555 timer, "normally on" means the output is high (voltage present) when the circuit is idle, while "normally off" means the output is low (no voltage) when idle. This depends on how the load is connected—either to Vcc or ground.

Due to Ohm's Law (V = IR), current flows from a higher voltage to a lower one. So, if the load is connected between output and ground, it turns on when the output is high (normally off). If the load is between Vcc and output, it turns on when the output is low (normally on).

Only one setup can be on at a time because the output pin can't be both high and low—it toggles between them, so when one setup is on, the other is off.

Astable Mode

Part 1

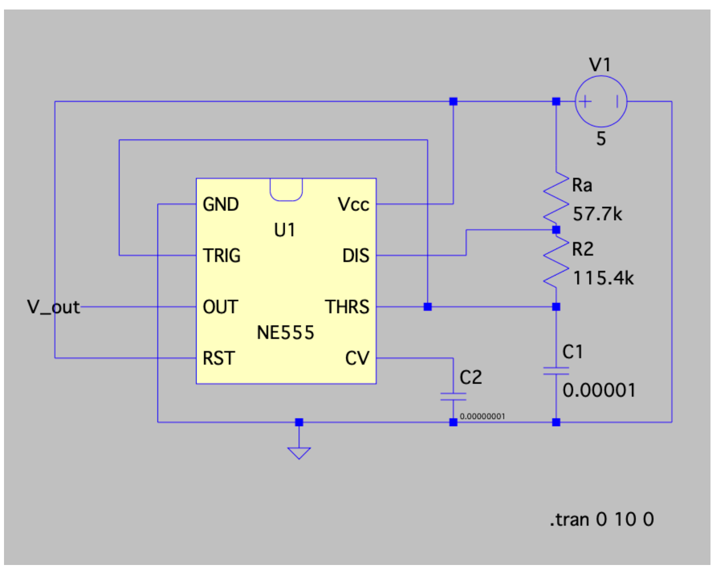


Fig2: Astable Schematic 1 with 60% duty cycle

- The aim was to create an astable circuit that has a duty cycle of 60% and a period of 2s.
- A capacitor of 10 micro-F and LM555 Timer were chosen for the task

- The required component values for the circuit were calculated as follows
    - $T = 0.693 \times (Ra + 2Rb) \times C$
    - Substitute the known values:
      $2 = 0.693 \times (Ra + 2Rb) \times (10 \times 10^{-6})$
    - Divide both sides by $0.693 \times 10 \times 10^{-6}$:
      $2 / (0.693 \times 10 \times 10^{-6}) = Ra + 2Rb$
      $2 / 6.93 \times 10^{-6} = Ra + 2Rb$
      $\sim 288{,}456.5 = Ra + 2Rb$
      (Equation 1)
      $Ra + 2Rb = 288{,}456.5$
    - Substitute D = 0.6:
      $0.6 = (Ra + Rb) / (Ra + 2Rb)$
    - Multiply both sides by $(Ra + 2Rb)$ to eliminate denominator:
      $0.6(Ra + 2Rb) = Ra + Rb$
      $0.6Ra + 1.2Rb = Ra + Rb$
      Bring terms to one side:
      $0.6Ra - Ra + 1.2Rb - Rb = 0$
      $-0.4Ra + 0.2Rb = 0$
      Divide entire equation by 0.2:
      $-2Ra + Rb = 0$
      (Equation 2)
      $Rb = 2Ra$
    - From Equation 2: $Rb = 2Ra$
      Substitute into Equation 1:
    - $Ra + 2(2Ra) = 288{,}456.5$
      $Ra + 4Ra = 288{,}456.5$
      $5Ra = 288{,}456.5$
      $Ra \approx 57{,}691.3$ ohms
    - Now use $Rb = 2Ra$:
      $Rb \approx 2 \times 57{,}691.3 \approx 115{,}382.6$ ohms
- The required circuit was constructed in LT Spice and the output value at V_out was graphed for testing.
- Ra is taken to be 56k ohms and Rb is taken to be 100k ohms for practical application.
- The circuit in the figure 2 was constructed with a 5V power supply.
- An oscilloscope was attached across the output pin of the timer and the common ground.
- The output data was observed on the oscilloscope.

Part 2

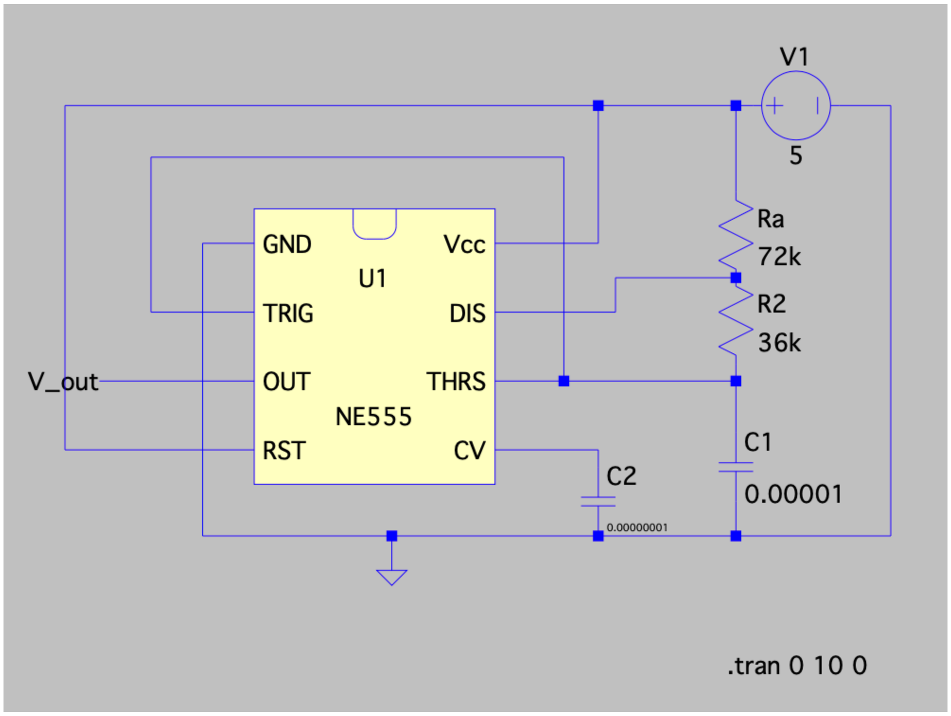


Fig3: Astable Schematic 2 with 75% duty cycle

- The aim was to create an astable circuit that has a duty cycle of 75% and a period of 1s.
- The required component values for the circuit were calculated as follows
  - `T = 0.693 × (Ra + 2Rb) × C`
  - `1 = 0.693 × (Ra + 2Rb) ×` (10×10^{-6})
  - Divide both sides:
  - $1 / (0.693 \times 10 \times 10^{-6}) = Ra + 2Rb$
    $1 / 6.93 \times 10^{-6} = Ra + 2Rb$
    `≈ 144,228.0 = Ra + 2Rb`
    (Equation 1)
    `Ra + 2Rb = 144,228.0`
  - Substitute D = 0.75:
  - `0.75 = (Ra + Rb) / (Ra + 2Rb)`
  - Multiply both sides by `(Ra + 2Rb)`:
  - `0.75(Ra + 2Rb) = Ra + Rb`
    `0.75Ra + 1.5Rb = Ra + Rb`
    Move all terms to one side:
  - `0.75Ra - Ra + 1.5Rb - Rb = 0`
    `-0.25Ra + 0.5Rb = 0`
  - Divide both sides by 0.25:
  - `-Ra + 2Rb = 0`
    (Equation 2)
    `Ra = 2Rb`
  - Substitute `Ra = 2Rb` into Equation 1:

- `2Rb + 2Rb = 144,228.0`
  `4Rb = 144,228.0`
  `Rb = 36,057.0 ohms`
- Then, `Ra = 2Rb = 72,114.0 ohms`

- The required circuit was constructed in LT Spice and the output value at V_out was graphed for testing.
- Ra is taken to be 68k ohms and Rb is taken to be 33k ohms for practical application.
- The circuit in the figure 3 was constructed with a 5V power supply.
- An oscilloscope was attached across the output pin of the timer and the common ground.
- The output data was observed on the oscilloscope.

Question: Instead of using fixed resistance and capacitor values to achieve different timing scenarios, use a variable component, such as a potentiometer to dynamically change the timing circuit in both monostable and astable mode. Where would this be helpful?

Using a potentiometer allows to dynamically adjust the timing without changing components. This is helpful in applications like LED dimmers, tone generators, or motor speed controls, where changing the pulse duration or frequency on the fly is needed for testing or user needs.

Application of 555 Timer – Electric Piano

Link: https://youtu.be/6dxMKBkFDPA

555 based Electric Piano:

This project simulates the fundamental operation of a digital musical instrument by using the 555 timer as a sound-generating oscillator. The circuit imitates a simple electric piano, where each key press (button) produces a distinct audio tone. This is achieved by varying the oscillation frequency of the 555 timer through different resistor combinations, selected via buttons.

Each pressed key connects a specific series of resistors into the 555 timer’s timing network, altering the RC time constant and thus the output frequency. The resulting square wave is sent to a small speaker, producing sound at frequencies corresponding to musical notes. The more resistors connected, the lower the frequency, allowing for tonal variation across the keyboard.

This setup reflects the core principle used in real-world synthesizers, electronic keyboards, and sound-generating toys, where oscillators produce variable frequencies mapped to user input. While basic, this circuit demonstrates how frequency control enables musical tone generation.

It serves as a practical example for understanding audio electronics and highlights the 555 timer’s role in low-cost sound applications, especially in educational tools and basic musical devices.

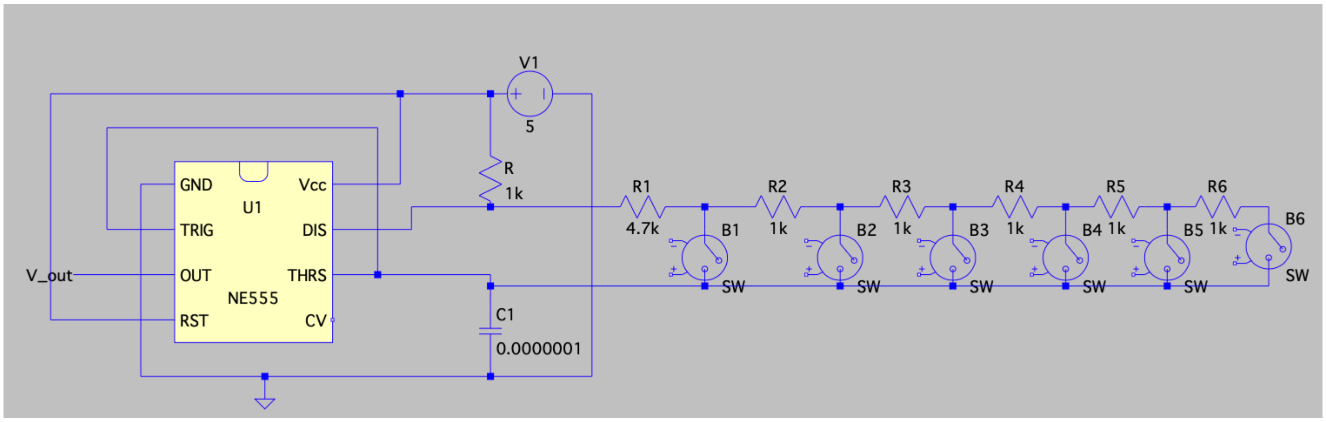


Fig4: Electric Piano Schematic

- The components chosen for the circuit include 6 x 1k ohm resistors and a 4.7k ohm resistor. The capacitor for the circuit was 0.1 micro-F
- The circuit was fully tested in LT Spice first to ensure proper frequency outputs for each button.
- The circuit was setup in an astable configuration, and a speaker was attached to the output pin of the 555 Timer.
- The complete circuit in figure 4 was constructed, wherein the switches were the buttons used as the keys of the piano.
- One of the 1k ohm resistor acted as Ra, while all the others connected in series acted as Rb (depending on the switch which is pressed).

Circuit operation

- The LM555 timer is configured in astable mode, meaning it continuously oscillates between high and low output states, generating a square wave at Pin 3 (OUT).
- The timing of these oscillations depends on the values of Ra, Rb, and C1.
- Ra is fixed at 1kΩ.
- Rb is variable and changes depending on which switch (B1–B6) is pressed.
- C1 is a fixed capacitor with a value of 0.1 µF (100nF).
- The right section of the circuit contains six 1kΩ resistors (R1–R6) and one 4.7kΩ resistor (R) connected in series.
- Each switch (B1– B6) taps the resistor chain at different points, effectively selecting the value of Rb.
- Pressing a switch closes a specific path, connecting a different number of resistors in series, which increases the overall Rb value from right to left.
- The 555 timer's frequency of oscillation is given by:
  - $f = 1.44 / ((Ra + 2 \times Rb) \times C)$
- As Rb increases, the denominator increases, resulting in a lower output frequency, which produces a lower-pitched tone from the speaker.

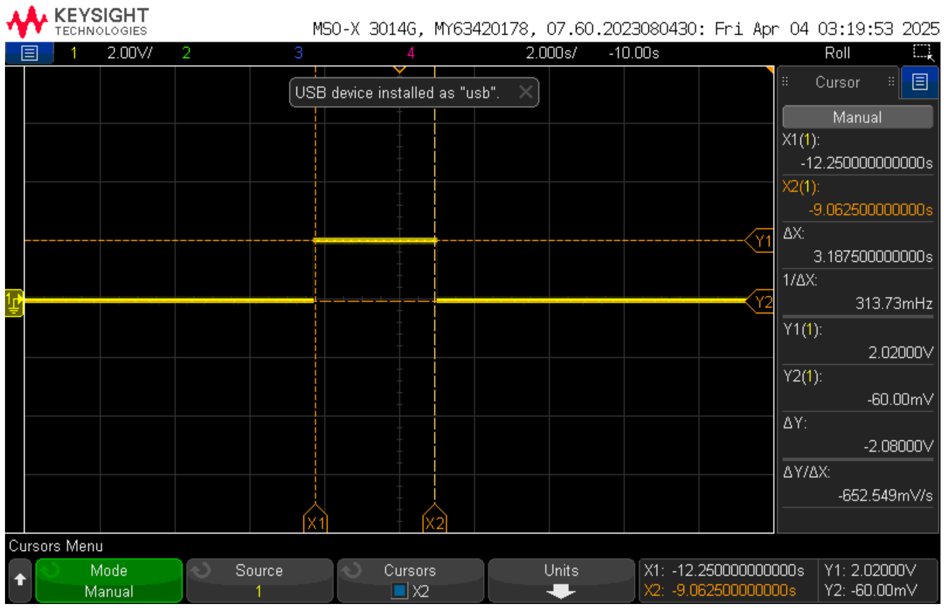
- Conversely, pressing a switch closer to the left connects fewer resistors in Rb, resulting in a higher frequency and higher-pitched tone.
- The output from Pin 3 is connected to a small speaker or buzzer.
- The square wave causes the speaker to vibrate at the selected frequency, producing an audible tone.
- Each switch press creates a distinct tone, similar to a piano key.

Results and Discussion

NOTE: All errors are calculated with respect to the theoretical (expected) values

Monostable Mode

Monostable simulation



Monostable Oscilloscope reading

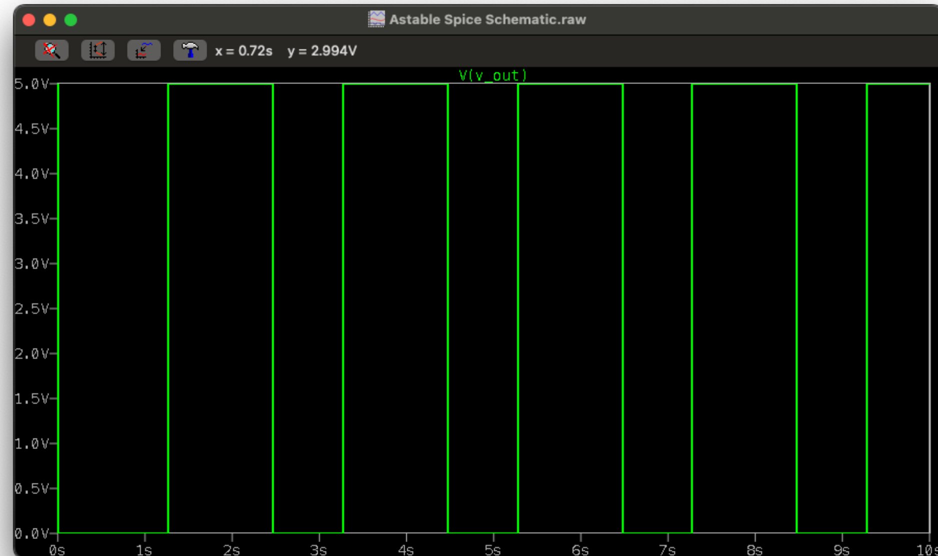
Table: Monostable high output pulse comparison between expected, simulated and measured values

Expected (s)	Simulated (s)	Measured(s)	Error %
3	3	3.187	6.23

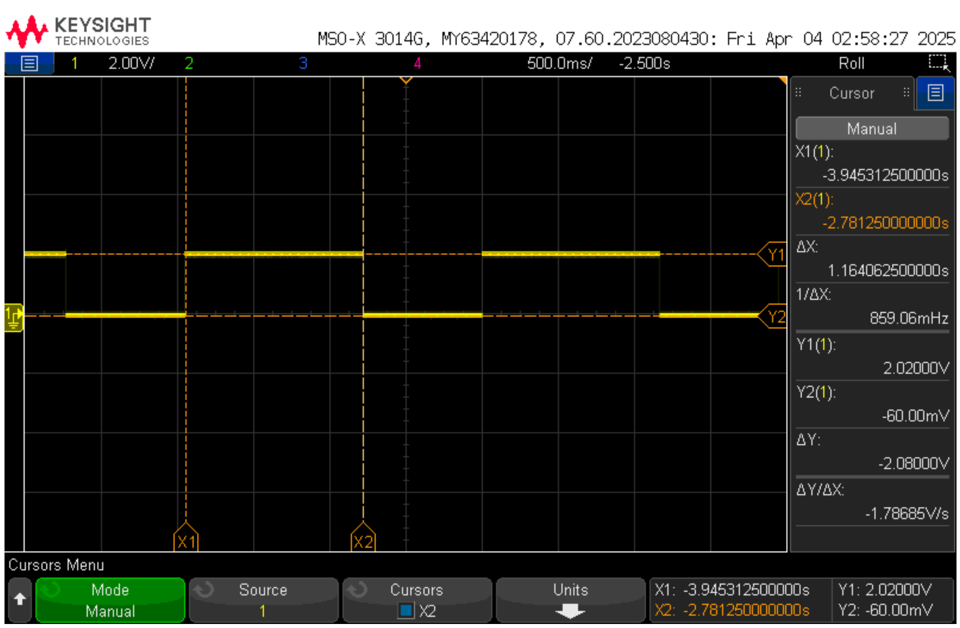
The measured pulse width of 3.187 seconds was slightly higher than the expected 3 seconds due to a combination of factors. The resistance used (276kΩ) was slightly higher than the calculated 273kΩ, directly increasing the time constant. Additionally, the 10 µF capacitor likely had a tolerance of ±10–20%, which could further extend the pulse duration. Variations due to temperature, component aging, or measurement precision may also have contributed. Minor inaccuracies in oscilloscope timing measurements may also contribute.

Astable Mode

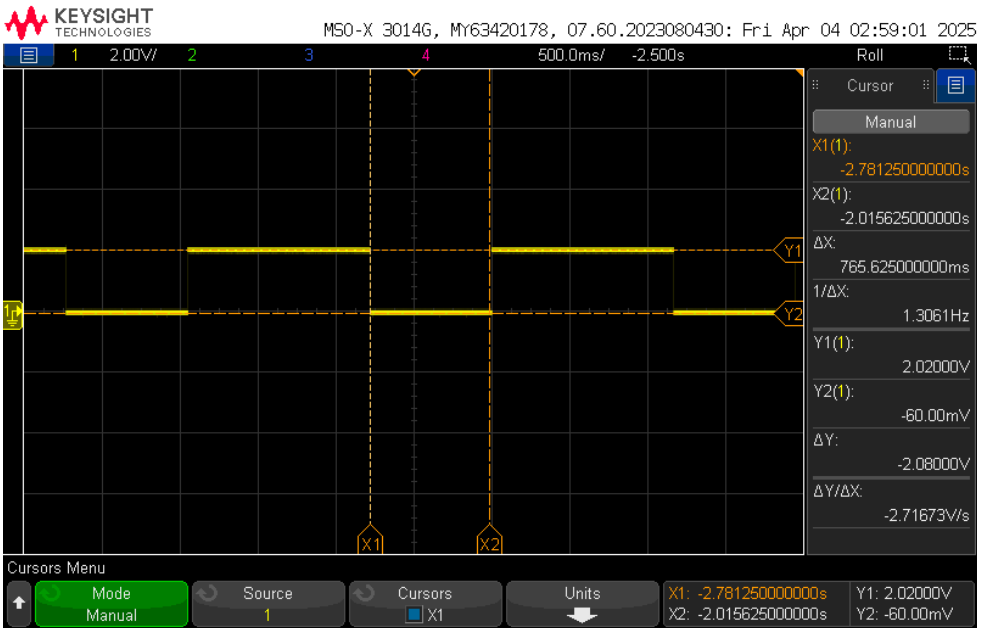
Part 1



Astable Simulation 1



Astable oscilloscope reading (high output)



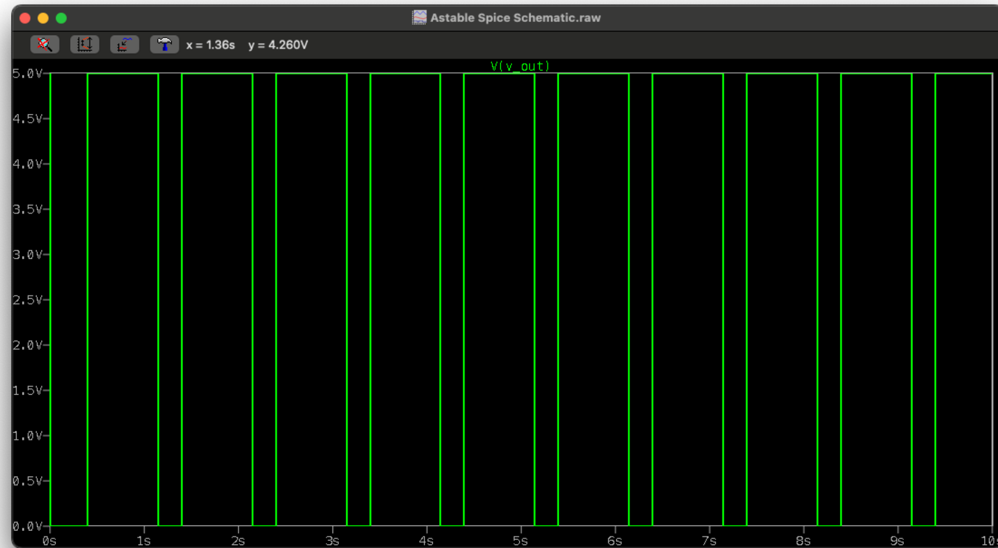
Astable oscilloscope reading (low output)

Table: Astable output pulse comparison between expected, simulated and measured values

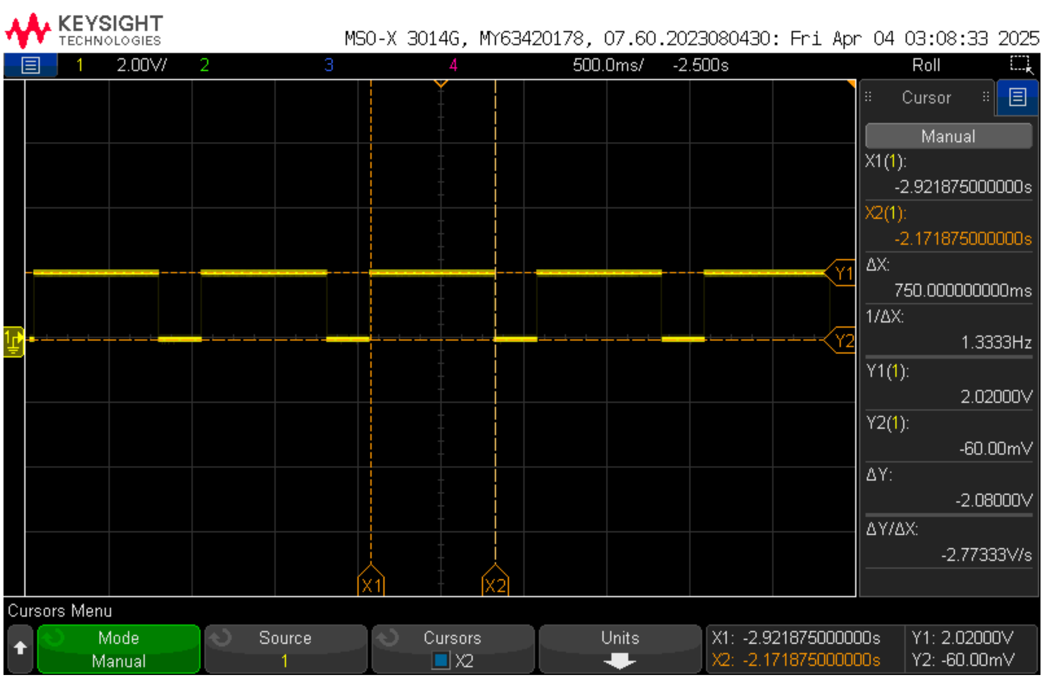
	Expected (s)	Simulated (s)	Measured (s)	Error %
High output	1.2	1.2	1.164	3
Low output	0.8	0.8	0.765	4.375

The measured high (1.164s) and low (0.765s) output durations deviated slightly from the expected values of 1.2s and 0.8s, with errors of 3% and 4.375%, respectively. These small deviations are primarily due to component tolerances. Additionally, the practical resistor values used (Ra = 56kΩ, Rb = 100kΩ) were slightly different from the calculated ones, affecting the RC time constant. Minor inaccuracies in oscilloscope timing measurements may also contribute.

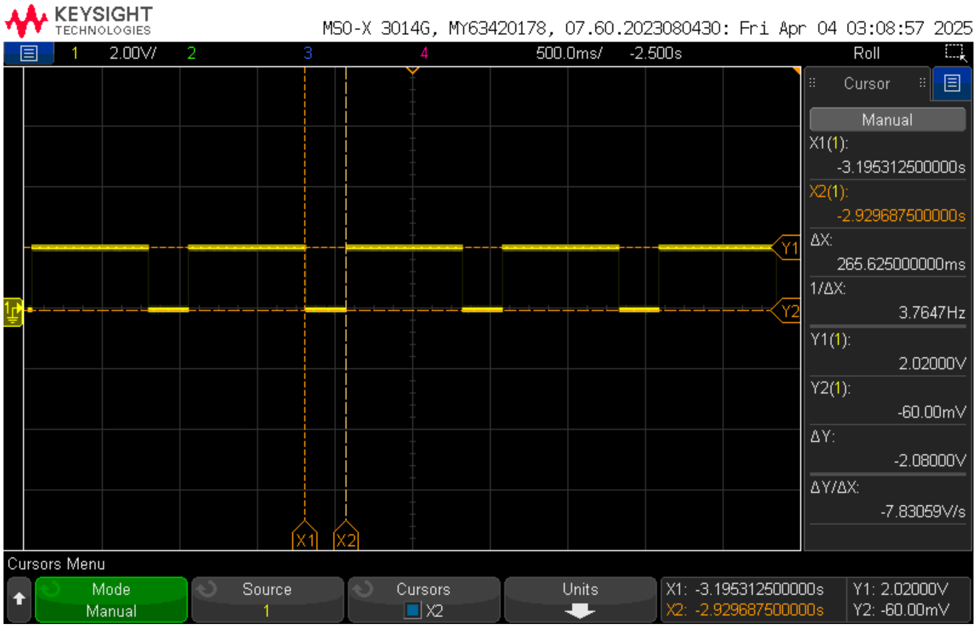
Part 2



Astable Simulation 2



Astable oscilloscope reading (high output)



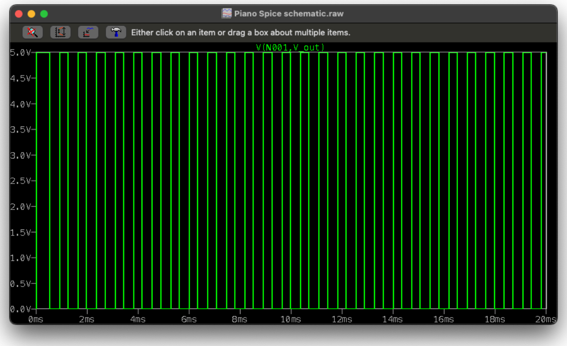
Astable oscilloscope reading (low output)

Table: Astable output pulse comparison between expected, simulated and measured values

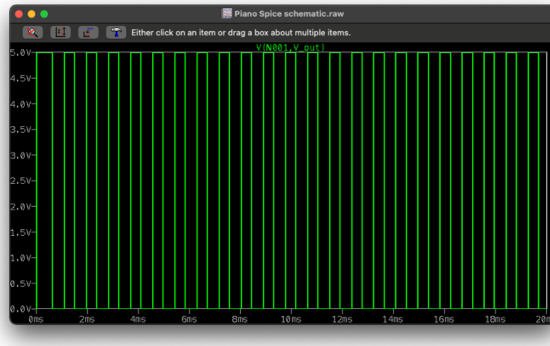
	Expected (s)	Simulated (s)	Measured (s)	Error %
High output	0.75	0.75	0.75	0
Low output	0.25	0.25	0.265	6

The measured low output duration (0.265s) showed a slight deviation from the expected 0.25s, resulting in a 6% error, while the high output matched perfectly. This deviation is likely due to standard component tolerances, particularly in the resistors and capacitor used, as real-world values may differ. Additionally, the practical resistor values (Ra = 68kΩ, Rb = 33kΩ) differ slightly from the ideal calculated values (Ra = 72.1kΩ, Rb = 36.1kΩ), which can affect the charge/discharge timing. Minor inaccuracies in oscilloscope timing measurements may also contribute.

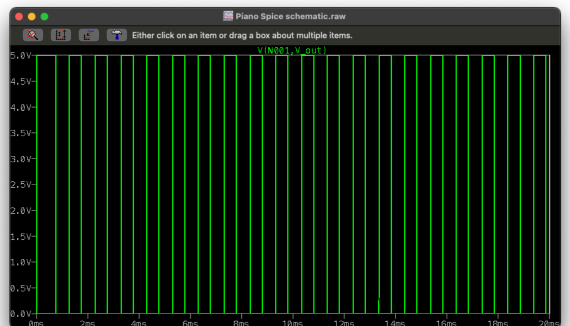
Application of 555 Timer – Electric Piano



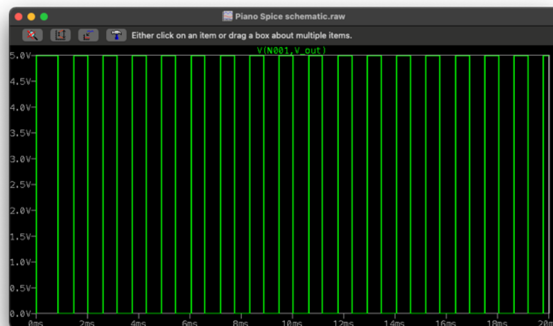
B1 toggled simulation



B2 toggled simulation

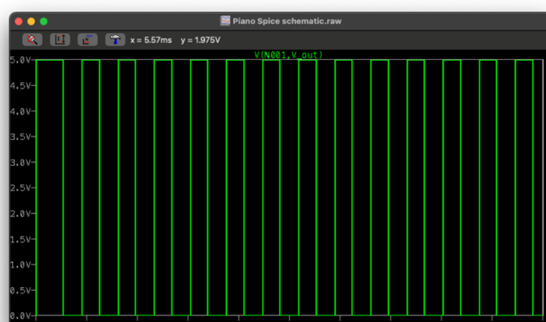


B3 toggled simulation

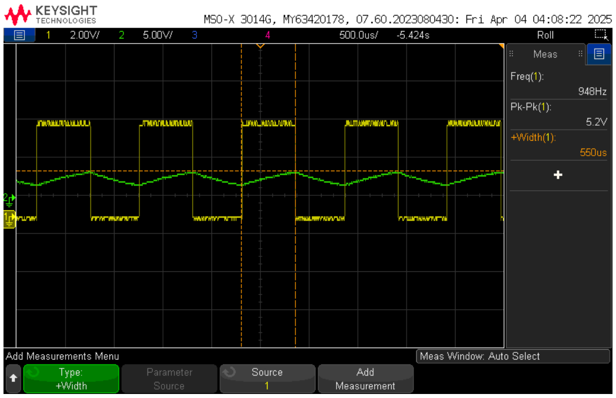


B4 toggled simulation

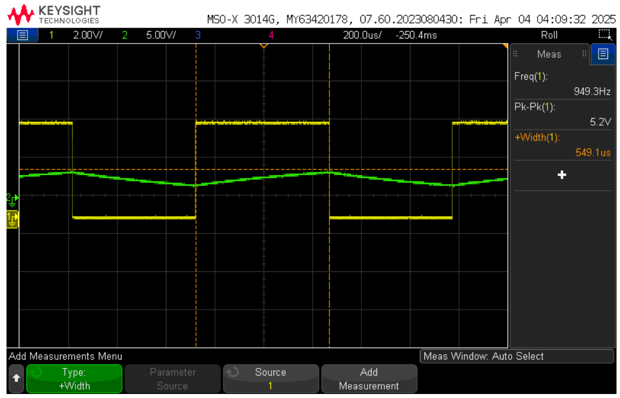
B5 toggled simulation



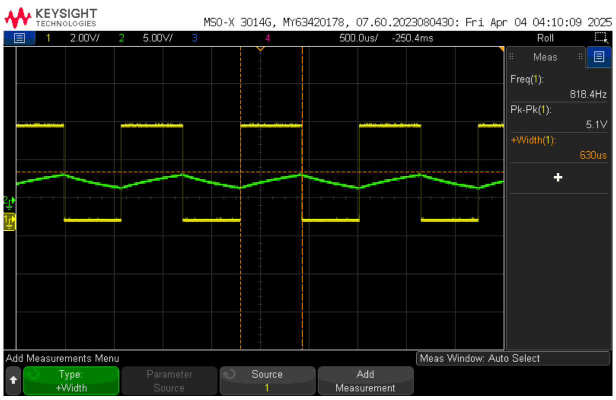
B6 toggled simulation



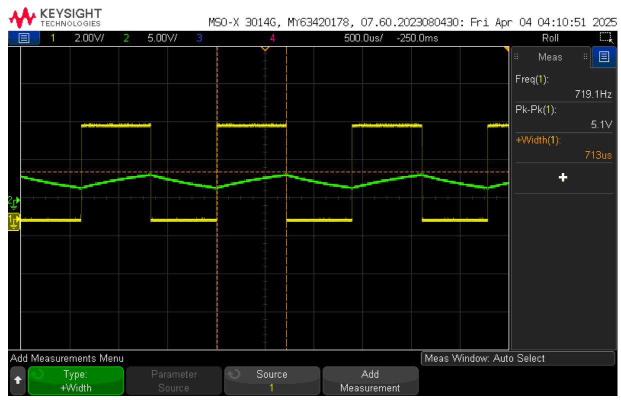
B1 toggled oscilloscope reading



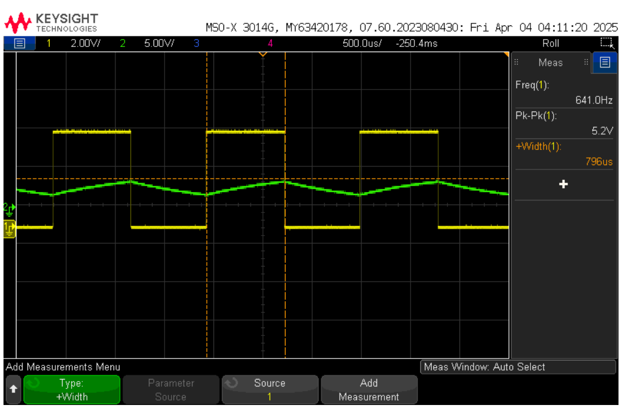
B2 toggled oscilloscope reading



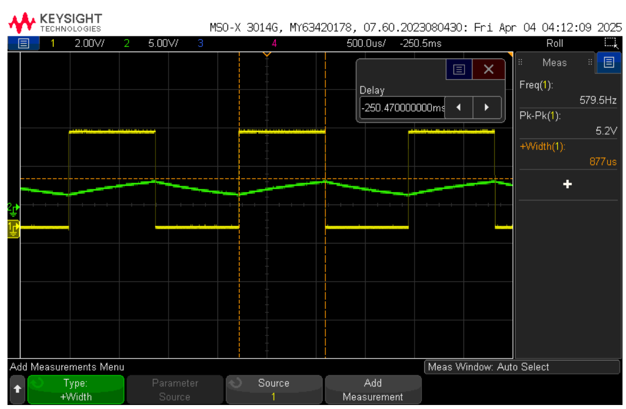
B3 toggled oscilloscope reading



B4 toggled oscilloscope reading



B5 toggled oscilloscope reading



B6 toggled oscilloscope reading

Table: Output frequency comparison between expected, simulated and measured values for all buttons

f = 1.44 / ((Ra + 2Rb) * C)

Expected (Hz)	Simulated (Hz)	Measured (Hz)	Error %
1385	1385	948	31.567
1161	1161	949.3	18.25
1000	1000	818.4	18.16
878	878	719.1	18.102
783	783	641	18.159
706	706	579.5	17.943

The measured frequencies showed noticeable deviations from the expected values, with errors ranging from 17.9% to 31.6%. These discrepancies are primarily due to component tolerances, especially in resistors and the 0.1 μF capacitor, which may vary significantly. Additionally, parasitic resistances from breadboard connections and switch contact resistance can affect the actual Rb seen by the timer. Oscilloscope resolution and rounding effects in frequency measurement can also contribute. Furthermore, the speaker itself consists of a capacitor and internal resistance, which may have led to the lower frequencies.

One important observation is that while the frequencies were lower than expected, the relative spacing between tones remained consistent. This means that the musical intervals between notes were preserved, resulting in tones that still sounded harmonically correct in sequence.

This consistency in frequency steps confirms that the circuit's design logic—based on incrementally increasing Rb through a series of resistors—functions as intended. Each additional 1kΩ resistor produced a predictable and proportional drop in frequency, preserving the relationship between consecutive notes even if the actual frequencies were slightly offset.

This insight demonstrates that even with component imperfections, the 555 timer can effectively mimic musical scales.

Conclusion

This project successfully demonstrated the functionality and versatility of the 555 timer in both monostable and astable configurations. The timing behavior matched theoretical expectations closely in most cases, and the electric piano application effectively showcased the timer's ability to generate variable audio tones using simple hardware.

Minor deviations were observed between expected and measured results, primarily due to component tolerances in resistors and capacitors and speaker load effects. Despite these, the relative behavior and waveform characteristics remained consistent.

Improvements such as using potentiometers for tuning, higher precision components, or replacing the resistor connections with a rotary switch could improve frequency accuracy. These changes would make the circuit more robust and suitable for applications in musical or signal generation.

Questions for deliverables:

1. Interest in this application

This application interests me because it combines my technical knowledge of electronics with my background in music as an experienced guitar player. Understanding how frequency and timing components like resistors and capacitors affect pitch allows me to explore the electronic generation of musical tones. Building a 555 timer-based piano circuit combines my passion for music with practical circuit design, and deepens my understanding of how signal shaping works in audio devices.

2. Pros and cons of design

Pros:

- Simple and low-cost design using only passive components and a 555 timer.
- No programming required, enabling standalone operation without a microcontroller.
- Real-time frequency control through physical resistor switching.
- Accurate tone stepping, preserving musical scales.

Cons:

- Frequency accuracy is affected by resistor and capacitor tolerances.
- Physical switches and wiring introduce parasitic resistance.
- Sound quality is basic, producing square waves without waveform shaping.
- Manual resistor selection introduces restrictions
- Speaker loading effect may distort output frequency.

3. Improvements and modifications

To better suit my interest as a musician, one practical enhancement would be adding a potentiometer in series with the speaker to serve as a volume control knob. This allows adjustment of the output amplitude without modifying the circuit or affecting the frequencies. It gives users the flexibility to control loudness based on preference or environment.

For more precise tonal control, the circuit can also be upgraded by replacing the resistor chain with a digital potentiometer, allowing smoother and more consistent frequency selection. This would enable better alignment with standard musical notes and scales.

4. Results matched or not

Monostable Mode:
The results were close to expected, with the measured pulse slightly longer due to component tolerances and minor resistance differences. Overall behavior matched theoretical predictions.

Astable Mode (60% & 75%):
The measured high and low pulses were slightly off but within acceptable error margins. Deviations were expected due to practical resistor values and capacitor tolerances.

Electric Piano:
The absolute frequencies were lower than expected, but the relative spacing between notes was preserved. While not perfectly tuned, the circuit behaved as intended and produced distinct tones per key.

5. What went well? What did not?

Circuit construction, simulations, and waveform measurements were successful across all experiments. The 555 timer behaved as expected in monostable and astable modes, and the electric piano produced distinct tones with consistent spacing.
Measured values deviated slightly due to component tolerances. The electric piano's absolute frequencies were lower than expected, and speaker loading may have affected output clarity.

Link: https://youtu.be/6dxMKBkFDPA

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